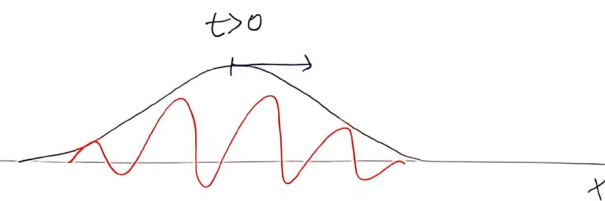
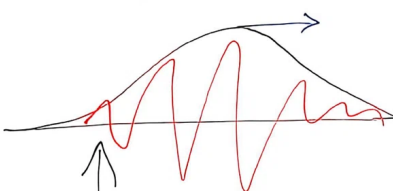


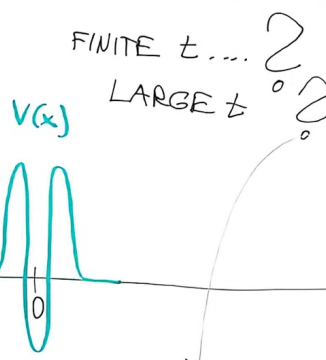
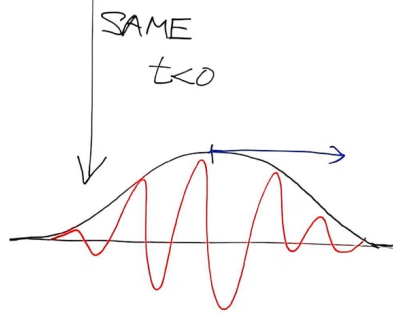
FREE WAVEPACKET
 $t < 0$



$\psi(0,x) \rightarrow \phi(p)$ GAUSSIAN

$$\psi(t,x) = \int_{-\infty}^{+\infty} dp \phi(p) \underbrace{\frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x}}_{\psi_p(x) \text{ FOR FREE PARTICLE}} e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}$$

FULL PROBLEM (WITH THE POTENTIAL)



APPENDIX III
 OF MY LECTURE
 NOTES

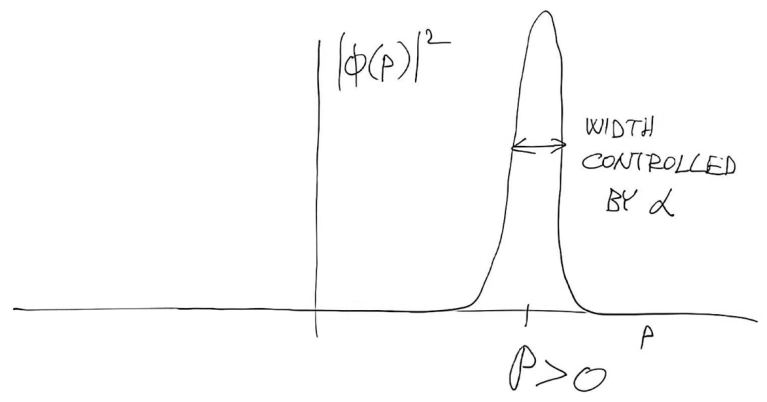
$$\psi(t,x) = \int_{-\infty}^{+\infty} dp \phi(p) \psi_p(x) e^{-\frac{i}{\hbar} \frac{p^2}{2m} t}$$

ASYMPTOTIC
 ANALYSIS

$R \rightarrow \infty$

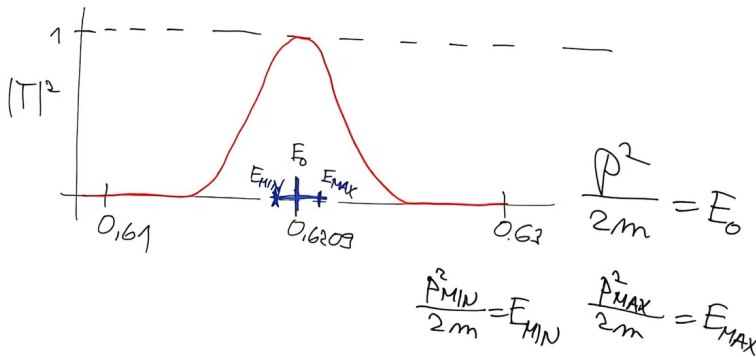
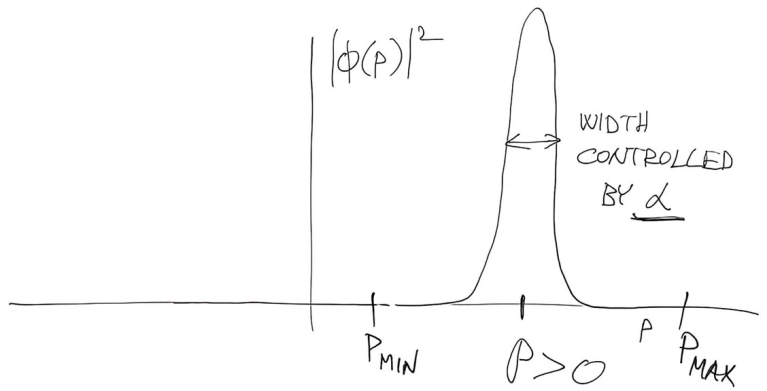
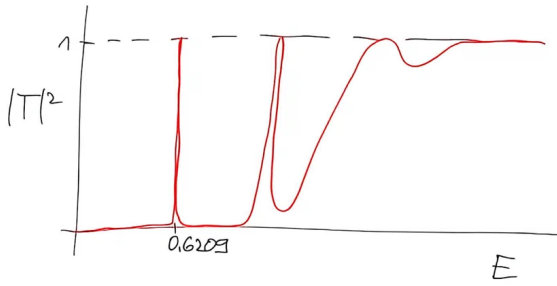
STATIONARY
 PHASE
 APPROXIMATION

$$\int_{-\infty}^{+\infty} d\tilde{z} \underbrace{f(\tilde{z})}_{\substack{\text{SMOOTH} \\ \subset \mathbb{R} \\ \text{COMPACT} \\ \text{SUPPORT}}} e^{+i R \tilde{z}}$$



$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\begin{aligned} \hat{H} \psi_p(x) &= E \psi_p(x) \quad ; \quad E = \frac{p^2}{2m} \\ \psi_p(x \rightarrow +\infty) &= T(p) \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} \\ \psi_p(x \rightarrow -\infty) &= \frac{1}{\sqrt{2\pi\hbar}} e^{+\frac{i}{\hbar} p x} \\ &\quad + R(p) \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} p x} \end{aligned}$$



$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\hat{H} \psi_p(x) = E \psi_p(x) \quad ; \quad E = \frac{p^2}{2m}$$

$$\psi_p(x \rightarrow +\infty) = T(p) \frac{1}{\sqrt{2\pi\hbar}} e^{+i\frac{p}{\hbar}x}$$

$$\psi_p(x \rightarrow -\infty) = \frac{1}{\sqrt{2\pi\hbar}} e^{+i\frac{p}{\hbar}x} + R(p) \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{p}{\hbar}x}$$

== END OF PROJECT. START OF THEORY. ==

FORMAL DEVELOPMENTS OF SCATTERING THEORY

SCATTERING IN THE TIME DOMAIN

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \quad V(x) \dots \text{SHORT RANGED}$$

TDSE

$$i\hbar \partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

$$i\hbar \partial_t \psi(t, x) = \hat{H} \psi(t, x)$$

FOR SIMPLICITY NOW!

NO BOUND STATES

ASYMPTOTIC CONDITION

(MATHEMATICAL THEOREM, APPENDIX III OF MY NOTES)

$t \rightarrow -\infty$

$$e^{-i\frac{\hat{H}}{\hbar}t} |\psi(0)\rangle$$

$$e^{-i\frac{\hat{H}_0}{\hbar}t} |\phi_{in}\rangle$$

in-STATE

0
 $t=0$
 $|\psi(0)\rangle$

$t \rightarrow +\infty$

$$e^{-i\frac{\hat{H}}{\hbar}t} |\psi(0)\rangle$$

$$e^{-i\frac{\hat{H}_0}{\hbar}t} |\phi_{out}\rangle$$

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle$$

$$\psi(t, x) = e^{-\frac{i}{\hbar} \hat{H} t} \psi(0, x)$$

TIME EVOLUTION
OPERATOR

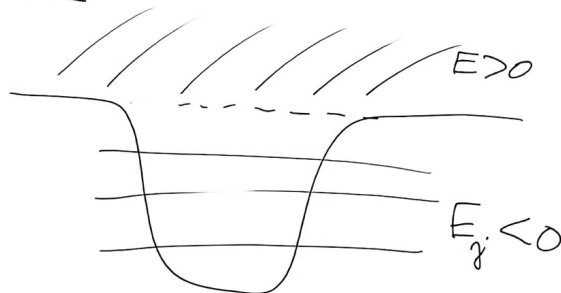
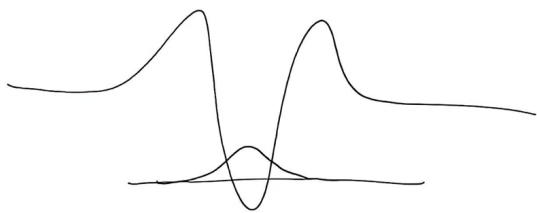
$$\begin{aligned} i\hbar \partial_t \left(e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle \right) &= \\ &= \cancel{i\hbar} \left(-\frac{i}{\hbar} \right) \hat{H} e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle = \\ &= \hat{H} |\psi(t)\rangle \end{aligned}$$

FOR ANY $|\psi(0)\rangle$ UNIQUE $|\phi_{in}\rangle$
..... UNIQUE $|\phi_{out}\rangle$

BASIC QUESTION OF SCATTERING
THEORY:

MAPPING $|\phi_{in}\rangle \mapsto |\phi_{out}\rangle$

THE CASE WHEN BOUND STATES EXIST:



$$|\phi_{out}\rangle = \sum |\phi_{in}\rangle$$

SCATTERING OPERATOR

FORMAL DEVELOPMENTS OF SCATTERING THEORY

SCATTERING IN THE TIME DOMAIN

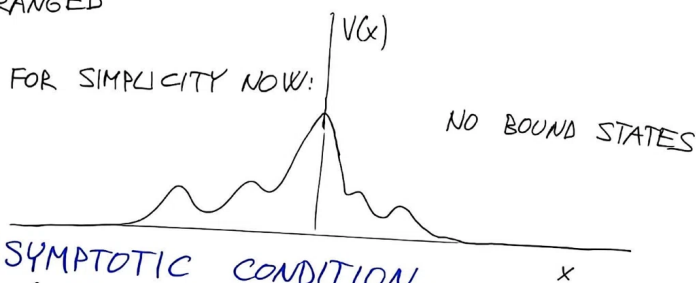
$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) \quad V(x) \dots \text{SHORT RANGED}$$

TDSE

$$i\hbar \partial_t |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

$$i\hbar \partial_t \psi(t, x) = \hat{H} \psi(t, x)$$

FOR SIMPLICITY NOW:



ASYMPTOTIC CONDITION

(MATHEMATICAL THEOREM, APPENDIX III OF MY NOTES)

$t \rightarrow -\infty$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle$$

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\hat{H}_0 = \frac{\hat{p}^2}{2m}$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle$$

in-STATE

0
 $t=0$

$|\psi(0)\rangle$
DOES NOT POPULATE
ANY BOUND STATES

$t \rightarrow +\infty$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle$$

$$e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{out}\rangle$$

LAST WEEK:

$$\psi(t, x) = \sum_j c_j \psi_j(x) e^{-\frac{i}{\hbar} E_j t} + \int_{-\infty}^{+\infty} dp \phi(p) \psi_p(x) e^{-\frac{i}{\hbar} E_m t}$$

FOR ALL t :

$$\langle \psi_j | \psi(t) \rangle = 0$$

$$e^A e^B \neq e^{A+B}$$

$\Downarrow$

$$e^{-\frac{i}{\hbar} (\hat{H}_0 - H) t} \text{ wrong}$$

$$\hat{A}(\lambda)$$

$$\lim_{\lambda \rightarrow \infty} \hat{A}(\lambda) = \hat{A}_\infty$$

STRONG CONVERGENCE

$$\lim_{\lambda \rightarrow \infty} \hat{A}(\lambda) |\psi\rangle = \hat{A}_\infty |\psi\rangle$$

WEAK CONVERGENCE

$$\lim_{\lambda \rightarrow \infty} \langle \psi_1 | \hat{A}(\lambda) | \psi_2 \rangle$$

$$= \langle \psi_1 | \hat{A}_\infty | \psi_2 \rangle$$

TAYLOR: SCATTERING THEORY (APPENDIX)

$|\phi_{in}\rangle$ GIVEN GET $|\phi_{out}\rangle$

$t \rightarrow -\infty$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle = e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{in}\rangle$$

$$|\phi_{in}\rangle = e^{+\frac{i}{\hbar} \hat{H}_0 t} e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle$$

LET $T > 0$ (AS LARGE AS WE WISH)

$$|\phi_{in}\rangle = e^{-\frac{i}{\hbar} \hat{H}_0 T} e^{+\frac{i}{\hbar} \hat{H} T} |\psi(0)\rangle$$

$$|\psi(0)\rangle = e^{-\frac{i}{\hbar} \hat{H} T} e^{+\frac{i}{\hbar} \hat{H}_0 T} |\phi_{in}\rangle$$

$$|\phi_{out}\rangle = \hat{S} |\phi_{in}\rangle$$

$t \rightarrow +\infty$

$$e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle = e^{-\frac{i}{\hbar} \hat{H}_0 t} |\phi_{out}\rangle$$

$$|\phi_{out}\rangle = e^{+\frac{i}{\hbar} \hat{H}_0 T} e^{-\frac{i}{\hbar} \hat{H} T} |\psi(0)\rangle \quad T \rightarrow +\infty$$

$T \rightarrow \infty$

$$|\phi_{out}\rangle = e^{+\frac{i}{\hbar} \hat{H}_0 T} e^{-\frac{i}{\hbar} \hat{H} T} e^{-\frac{i}{\hbar} \hat{H} T} e^{+\frac{i}{\hbar} \hat{H}_0 T} |\phi_{in}\rangle$$

$$|\phi_{out}\rangle = e^{+\frac{i}{\hbar} \hat{H}_0 T} e^{-2\frac{i}{\hbar} \hat{H} T} e^{+\frac{i}{\hbar} \hat{H}_0 T} |\phi_{in}\rangle$$

$T \rightarrow +\infty$

$$\hat{S}$$

$$\hat{S}^+ = \hat{S}^{-1}$$

$$e^{-\frac{i}{\hbar} \hat{H} T} e^{\frac{i}{\hbar} \hat{H}_0 T} |\phi_{in}\rangle = |\psi(0)\rangle$$

SIMON & REED

$$\hat{U}^\dagger(-T) \hat{U}_0(-T) |\phi_{in}\rangle$$

$$\hat{U}(t) = e^{-\frac{i}{\hbar} \hat{H} t}$$

$$\hat{U}_0(t) = e^{-\frac{i}{\hbar} \hat{H}_0 t}$$

$t \rightarrow -\infty$

$$\hat{U}^\dagger(t) \hat{U}_0(t) |\phi_{in}\rangle = |\phi_{in}\rangle - \int_t^0 dt' \frac{d}{dt'} \left\{ \hat{U}^\dagger(t') \hat{U}_0(t') |\phi_{in}\rangle \right\}$$

$$\hat{U}^\dagger(t') \hat{U}_0(t') |\phi_{in}\rangle \Big|_0^t$$

$$= |\phi_{in}\rangle + \left(\hat{U}^\dagger(t') \hat{U}_0(t') |\phi_{in}\rangle \Big|_t^t \right)$$

$$= |\phi_{in}\rangle + \hat{U}^\dagger(t) \hat{U}_0(t) |\phi_{in}\rangle = \hat{U}^\dagger(0) \hat{U}_0(0) |\phi_{in}\rangle$$